

Aakash Siricilla

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years designing, building, and operating distributed backend systems in production. Builds high-throughput Java/Spring Boot and Python/FastAPI services and event-driven Kafka pipelines on AWS, and owns them end to end: API and schema design, Kubernetes deployment, CI/CD, observability, on-call, and incident root-cause analysis. Also ships production ML services and LLM agents, including TARS, an autonomous computer-use agent backed by ~1,200 tests.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, SQL, Bash, C/C++

Backend: Spring Boot, FastAPI, REST, gRPC, WebSockets, Microservices, Kafka, Celery, Event-Driven Architecture

Data & Storage: PostgreSQL, MySQL, Redis, TimescaleDB, SQLite, Query Optimization, Schema Design, ETL/Streaming Pipelines

Cloud & DevOps: AWS (EC2, EKS, S3, RDS, Lambda, SQS, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, Git, Linux

Observability & Testing: Prometheus, Grafana, CloudWatch, SLOs/SLIs, JUnit, Mockito, pytest, k6

Frontend: React, Next.js, Tailwind CSS

AI/ML: PyTorch, scikit-learn, XGBoost, MLflow, LLMs, RAG, Vector Search (Qdrant, FAISS)

PROFESSIONAL EXPERIENCE

AT&T Sep 2024 – Present
Software Engineer, Distributed Systems *Remote, USA*

- Designed and built an event-driven service-assurance platform (Java, Spring Boot, Kafka, PostgreSQL) ingesting ~40M network and customer-operations events/day from 12+ upstream systems, replacing batch reconciliation and cutting manual effort 40%.
- Built REST/gRPC APIs sustaining ~2.5k RPS with connection pooling, Redis caching, and idempotent writes; rewrote hot-path PostgreSQL queries with composite indexes and query-plan tuning, cutting p95 latency 35% (480ms → 310ms).
- Deployed services across dev/stage/prod on AWS EKS with Docker, Terraform, and GitHub Actions CI/CD, adding canary releases, automated rollbacks, and config management that took release-related incidents from ~3/month to near zero.
- Built a Python/FastAPI anomaly-detection service (scikit-learn, XGBoost) that scores streaming operational metrics against seasonal baselines, with MLflow-tracked retraining and drift monitoring; cut false pages 30% and time-to-detect from hours to under 10 minutes.
- Defined SLOs/SLIs (99.9% availability, p95 latency) with burn-rate alerts in Prometheus, Grafana, and CloudWatch, and wrote the runbooks; serve in the on-call rotation and lead P1/P2 root-cause analysis.
- Automated health checks, log parsing, data validation, backfills, and scheduled maintenance in Python and Bash, eliminating ~15 hours/week of manual runbook work and reducing on-call toil.
- Delivered role-based access control and transactional audit logging, cutting investigation time 45%; partnered with product, network operations, and security teams on requirements, capacity plans, and test procedures.

Flipkart Apr 2021 – Sep 2023
Software Engineer *Hyderabad, India*

- Built CRM and workflow-automation services (Java, Spring Boot, React, PostgreSQL) used by 1,500+ internal agents for lead management and client onboarding, reducing manual coordination effort 35%.
- Designed REST APIs, service-layer validation, and data-access logic for transaction-heavy flows handling ~5M records/day; added Redis caching and removed N+1 queries, improving retrieval performance 25% and holding p99 latency within target through festive-sale peaks.
- Built asynchronous job processing (Kafka consumers and scheduled workers) for notifications, exports, and batch updates, moving long-running work off request threads and tripling job throughput with retries and dead-letter handling.
- Built monitoring dashboards and alerts for API health, queue lag, and error rates; resolved production issues within service-level objectives and shipped fixes through CI with JUnit/Mockito coverage above 80%.
- Created a reusable React component library integrated with Spring Boot services, cutting feature delivery time 30% and standardizing UI across four teams.

PROJECTS

TARS – Autonomous LLM Computer-Use Agent | *Python, FastAPI, Chrome MV3, Electron, SQLite* GitHub

- Architected an agent that completes multi-step tasks in web and desktop apps through a screenshot → model action → execute → read-back verification loop, with swappable execution backends: a DOM-verified Chrome extension and a desktop driver (UIA/SendInput).
- Built the reliability and safety layer: approval gates before any submit or payment, a credential vault the model never sees, sensitive-field masking in screenshots, URL allow-listing, and per-step receipts, on a FastAPI/SQLite backend covered by ~1,200 tests.

SyncStream – Real-Time Collaborative Streaming | *Java, Spring Boot, Next.js, WebSockets, PostgreSQL, Docker* GitHub

- Built a real-time collaboration platform with invite-based sessions, persistent room state, and low-latency WebSocket sync between users; containerized frontend and backend with Docker.

DocuMindAI – Document Intelligence (RAG) Platform | *FastAPI, React, Qdrant, Redis, Celery, MLflow, Docker* GitHub

- Built a document platform with async ingestion, chunking/embedding pipelines, hybrid (dense + keyword) retrieval over Qdrant, and LLM chat; offloaded indexing to Celery/Redis workers and tracked retrieval experiments in MLflow, cutting lookup time ~70%.

EDUCATION

New Jersey Institute of Technology, Master of Science in Computer Science Sep 2023 – May 2025
Coursework: Distributed Systems, Deep Learning, Machine Learning & MLOps **GPA: 3.93/4.0**

Kommuri Pratap Reddy Institute of Technology, B.Tech. in Computer Science & Engineering Aug 2018 – May 2022